

Modeling Dissipative Kerr Solitons in Optical Cavities with the Lugiato-Lefever Equation

Alessandro Di Marzo

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Abstract

This project investigates the temporal Lugiato-Lefever Equation (LLE). The LLE is a non-linear partial differential equation modeling the evolution of the intracavity field in a Kerr medium optical resonator. We derive the LLE via the nonlinear Schrödinger equation (NLSE), incorporating Kerr nonlinearity and group velocity dispersion under the slowly varying envelope approximation (SVEA). We implement a numerical solution to the dimensionless LLE using the Split-Step Fourier Method (SSFM) and second order Runge-Kutta integration for better efficiency implemented in Python. The simulation explores the formation of dissipative Kerr solitons and their stability under varying detuning and driving parameters. The simulations generate single soliton, multiple soliton and soliton crystal states validating the SSFM approach as a tool for modeling nonlinear optical cavity dynamics. Python code and animations of the simulation can be found in this GitHub repository.

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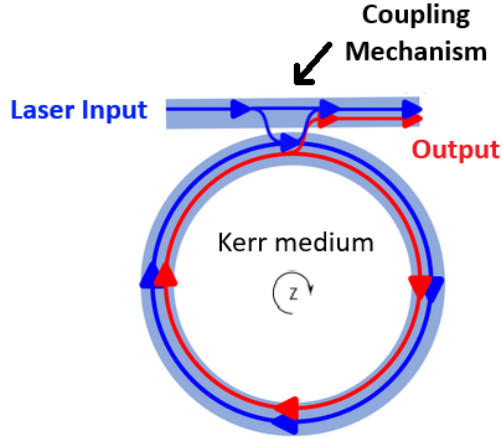


Figure 1: Simple schematic of a Kerr ring resonator. In the spatial version of the LLE, the z axis goes along the path of the ring. The coupling mechanism has reflection (R) and transmission (T) coefficients such that $T + R = 1$. In optical microresonators, light is trapped by circulating it around the ring under the right conditions with the goal of generating dissipative Kerr solitons.

1 The Lugiato-Lefever Equation

This section introduces the physical context, historical development, and theoretical foundations of the Lugiato-Lefever equation (LLE), starting from the nonlinear Schrödinger equation (NLSE) and culminating in the temporal form used in this project. Understanding the derivation and structure of the LLE is essential before implementing numerical methods to solve it in section 2. The main goal when solving the the LLE is to generate dissipative Kerr solitons (DKS) which are localized pulses of light that circulate inside the resonator and, thanks to a balance between dispersion, Kerr nonlinearity, cavity loss, and external driving, remain stable over time. When these solitons circulate, they create a very regular train of pulses in time and comb-like lines in frequency space. These combs can act as precise “rulers” in the frequency domain that are used for optical clocks [1], spectroscopy [2] and other applications.

The LLE is a nonlinear partial differential equation describing the evolution of the field envelope inside an optical cavity such as the one shown in figure 1. It can be classified as a driven, damped, detuned nonlinear Schrodinger equation. The medium inside the cavity exhibits Kerr nonlinearity (more on this in section 1.2.1). The spatial transverse LLE takes the form:

$$\frac{\partial E(z)}{\partial z} = [-1 - i(\Delta - |E|^2) + i\nabla_{\perp}^2] E + F \quad (1)$$

where $E(z)$ is the slowly varying envelope of the electric field inside the cavity, Δ is the cavity detuning, and F is the normalized input field. z is the distance along the path of the ring. Note that the LLE is describing the envelope of the carrier wave rather than the intensity of the optical field itself. This is known as the slowly varying envelope approximation (SVEA) and it is valid because the carrier wave is changing much faster than the envelope (more on this in section 1.2). That being said, the main focus of this project is the temporal longitudinal version of the LLE given by:

$$\frac{\partial E(t, \tau)}{\partial t} = \left[-1 - i(\Delta - |E|^2) + i \frac{\partial^2}{\partial \tau^2} \right] E + F \quad (2)$$

Here, the LLE has no spatial derivatives and has an extra independent variable $\tau = t - \frac{z}{v_g}$ where v_g is the group velocity. τ can be thought of as a time window that “follows” a section of the ring as t evolves. From here onward, τ will be referred to as the fast time and t as the slow time. We have also assumed conditions for anomalous dispersion (more on this in section 1.2).

1.1 Historical Background

The LLE was introduced in 1987 by Luigi Lugiato and René Lefever in their paper titled “*Spatial dissipative structures in passive optical systems*”[3]. The equation was developed to describe the formation of spatial patterns in optical cavities with a Kerr medium, driven by continuous wave (CW) lasers. The LLE was designed as an optical analog to the Brusselator model [4] in chemical pattern formation. In chemical systems, the Brusselator models Turing patterns caused by nonlinear reactions and diffusion. In optics, the LLE plays a similar role but with diffraction replacing diffusion, and a Kerr medium as the source of nonlinearity. It emerged from simplifications of the Maxwell–Bloch equations [5] for a cavity filled with a nonlinear medium. Through adiabatic elimination and under the high-Q cavity (low loss) and single mode approximations, the model was reduced to a partial differential equation [5] [6]. Initially, the LLE was used to predict and analyze spatial patterns such as rolls and hexagons in transverse cavity geometries. In the following years, researchers formulated the temporal version of the LLE that explores patterns in the longitudinal direction (z-direction) [7]. We can see the difference by comparing equations 1 and 2. The main difference being the introduction of a dispersion term instead of a diffraction term. The diffraction term that depended on the second derivatives of the transverse variables were replaced by the second derivative with respect to the fast time τ that “moves” with the pulse as slow time t evolves. This version now describes patterns in the z direction while the intensity in the x and y directions remain uniform.

1.2 The Nonlinear Schrodinger Equation

In this section, we briefly outline the relevant theory of nonlinearity and dispersion which are important in understanding the generalized version of the NLSE.

1.2.1 Nonlinearity

Nonlinear optics studies how light behaves and interacts in materials where the response (polarization) of the medium is not proportional to the strength of the electric field of the light. Unlike linear optics, where the medium's response scales directly with light intensity, in nonlinear optics the optical properties like the refractive index of the material change in ways that are not simply proportional to the input [8]. This behavior is usually observed at high light intensities, like those created by lasers. In nonlinear optics, we can describe the induced polarization in the material $P(t)$ by

$$P(t) = \epsilon_0 (\chi^{(1)} E(t) + \chi^{(2)} E(t)^2 + \chi^{(3)} E(t)^3 + \dots) \quad (3)$$

where ϵ_0 is the permittivity in free space and $E(t)$ is the applied optical field that is causing the response. The $\chi^{(n)}$ coefficients are the n^{th} order susceptibilities of the medium which are tensors that describe how $P(t)$ responds to $E(t)$. This project is concerned with a special kind of nonlinear material known as a Kerr material [9]. Since the Kerr material displays inversion symmetry, the second order term becomes zero [8]. From here, we make several assumptions about the materials in the setup introduced in figure 1. Firstly, we assume that the fourth order and higher terms are small enough to be ignored. Also, we assume that the polarization changes instantaneously in response to the applied optical field. Eliminating the second order term and the terms of fourth and higher order, equation 3 becomes

$$P(t) = \epsilon_0 (\chi^{(1)} E(t) + \chi^{(3)} E(t)^3) = P_L + P_{NL} \quad (4)$$

where we are left with two terms: the linear polarization P_L and the nonlinear polarization P_{NL} .

1.2.2 Dispersion

Chromatic dispersion refers to the frequency dependence of a medium's refractive index, denoted as $n(\omega)$, and commonly arises in the interaction of light with dielectric media like silica (which is a common material for fiber optic cables). This phenomenon causes different spectral components of a light pulse to propagate at different velocities, resulting in temporal broadening or "chirping" of the pulse. To simplify the modeling of dispersion, we employ the Slowly Varying Envelope Approximation (SVEA). This approximation assumes that the envelope of the optical

field varies much more slowly than its carrier wave in both space and time:

$$\left| \frac{\partial E}{\partial z} \right| \ll \beta_0 |E|, \quad \left| \frac{\partial E}{\partial t} \right| \ll \omega_0 |E| \quad (5)$$

Under this approximation, the propagation constant $\beta(\omega)$ can be Taylor expanded around the carrier frequency ω_0 :

$$\beta(\omega) \approx \beta_0 + \beta_1(\omega - \omega_0) + \frac{1}{2}\beta_2(\omega - \omega_0)^2 \quad (6)$$

where

$$\beta_0 = \beta(\omega_0), \quad \beta_1 = \left. \frac{\partial \beta}{\partial \omega} \right|_{\omega_0} = \frac{1}{v_g}, \quad \beta_2 = \left. \frac{\partial^2 \beta}{\partial \omega^2} \right|_{\omega_0} \quad (7)$$

Here, v_g is the group velocity, and β_2 characterizes the group velocity dispersion. Applying the inverse Fourier transform to the non-constant terms of the expansion yields:

$$\mathcal{F}^{-1} \left(\beta_1(\omega - \omega_0) + \frac{1}{2}\beta_2(\omega - \omega_0)^2 \right) = i\beta_1 \frac{\partial E}{\partial t} - \frac{\beta_2}{2} \frac{\partial^2 E}{\partial t^2} \quad (8)$$

This result, when combined with the nonlinear polarization from Kerr effects, contributes to the derivation of the NLSE, which serves as a foundation for the Lugiato-Lefever Equation in Kerr ring resonators.

1.2.3 Generalized NLSE

The NLSE can be derived from the wave equation for the electric field $\vec{E}$ that comes from Maxwell's equations. For more details, see [8] or [10]. The result is the propagation equation for an envelope $A(z, t)$:

$$\frac{\partial A}{\partial z} + \beta_1 \frac{\partial A}{\partial t} + \frac{i\beta_2}{2} \frac{\partial^2 A}{\partial t^2} = i\gamma |A|^2 A \quad (9)$$

where β_1 and β_2 represent the first and second order dispersion parameters and γ quantifies the nonlinear interaction strength. This equation combines the effect of Kerr nonlinearity from equation 4 and the dispersion effect of equation 8. The full derivation for the NLSE can be found at [5][8][11].

The next step is to transform to a reference frame moving at the group velocity $v_g = 1/\beta_1$. The equation becomes:

$$\frac{\partial E_n}{\partial z} = -i\frac{\beta_2}{2} \frac{\partial^2 E_n}{\partial \tau^2} + i\gamma |E_n|^2 E_n \quad (10)$$

which is the generalized nonlinear Schrödinger equation. We have also done some relabeling to better describe a ring resonator setup. $E_n(z, \tau)$ is the envelope of the field on the n^{th} roundtrip around the ring and $\tau = t - z/v_g$ is the fast time.

1.2.4 The Temporal LLE

The next step is to obtain the temporal LLE from the NLSE above. More details can be found in the text where it was first derived by Halterman, Trillo, and Wabnitz [7] or in [12]. The idea is to apply the following boundary condition at the coupling mechanism ($z = 0$) to the generalized NLSE:

$$E_{n+1}(0, \tau) \approx \sqrt{T}F + \sqrt{R}e^{-i\Delta}\mathcal{G}_L E_n(0, \tau) \quad (11)$$

where T and R are the transmission and reflection coefficients of the coupling mechanism in figure 1 respectively. Δ is the cavity detuning. The detuning is the difference between the input laser's frequency and the closest resonant frequency in the cavity. It plays a significant role in pattern generation with the LLE which we will see more in section 3. F is the input field intensity and the term $\sqrt{T}F$ refers to its effectiveness at the coupling mechanism. Lastly, $\mathcal{G}_L$ is an operator that accounts for the pulse propagation over one roundtrip around the ring with circumference L .

$$\mathcal{G}_L E_n(0, \tau) = \int_0^L \frac{\partial E_n}{\partial z} dz \quad (12)$$

The integrand is nothing other than the NLSE from equation 10 and the integral is solved by using the mean field approximation which is valid for low loss cavities [3][5]. In the case where $T \ll 1$ (low loss), a photon in the cavity stays within it for a long time compared to the roundtrip time t_R . In other words, few photons are leaving and entering the cavity and we can say that the intracavity field intensity changes little over a single roundtrip [10]. Therefore it will suffice to say that the change in E over a roundtrip is the rate of change at the beginning ($z = 0$) multiplied by the total distance L . Then, we can add the field strength at the coupling with this change to get a good estimate of how much this function changes over the whole ring.

$$\mathcal{G}_L E_n(0, \tau) \approx E_n(0, \tau) + L \frac{\partial E_n(0, \tau)}{\partial z} \quad (13)$$

plugging in equation 10,

$$\mathcal{G}_L E_n(0, \tau) = E_n(0, \tau) + L \left(-i \frac{\beta_2}{2} \frac{\partial^2 E(0, \tau)}{\partial \tau^2} + i\gamma |E_n(0, \tau)|^2 E_n(0, \tau) \right) \quad (14)$$

This will eventually be substituted back into the boundary condition (equation 11). Before this however, there are simplifications that can be made to equation 11. We approximate $\sqrt{R} \approx 1 - \frac{T}{2}$ for low loss ($T \ll 1$), and use the expansion $e^{-i\Delta} \approx 1 - i\Delta$. Then we can write:

$$E_{n+1}(0, \tau) \approx \sqrt{T}F + \left(1 - \frac{T}{2} - i\Delta + i\frac{T\Delta}{2} \right) \mathcal{G}_L E_n(0, \tau) \quad (15)$$

To simplify further, we use the assumption that cavity detuning Δ and the nonlinear effect $L\gamma E$ are small ($\ll 1$). We say the dispersion and nonlinearity terms in equation 14 are of first order smallness [7] and any higher order terms can be neglected when combining equation 15 and 14. Plugging in equation 14 into 15 with this assumption gives:

$$E_{n+1}(0, \tau) \approx \sqrt{T}F + \left(1 - \frac{T}{2} - i\Delta - iL\frac{\beta_2}{2}\frac{\partial^2}{\partial\tau^2} + iL\gamma|E(0, \tau)|^2\right) E(0, \tau) \quad (16)$$

To write this with a continuous time variable rather than with n discrete steps, slow time is defined as $t = nt_R$. This means $E_n(z = 0, \tau) = E(t = nt_R, \tau)$. Furthermore, its derivative is:

$$\frac{\partial E(t=nt_R, \tau)}{\partial t} = \frac{E_{n+1}(0, \tau) - E_n(0, \tau)}{t_R} \quad (17)$$

$$E_{n+1}(0, \tau) = t_R \frac{\partial E(t=nt_R, \tau)}{\partial t} + E_n(0, \tau)$$

Inserting the second part of equation 17 into equation 16 gives

$$t_R \frac{\partial E(t, \tau)}{\partial t} = \sqrt{T}F + \left[-\frac{T}{2} - i\Delta - iL\frac{\beta_2}{2}\frac{\partial^2}{\partial\tau^2} + iL\gamma|E(t, \tau)|^2\right] E(t, \tau) \quad (18)$$

Equation 18 is referred to as the dimensional LLE and with appropriate rescaling:

$$\begin{aligned} \alpha = \frac{T}{2}, \quad \bar{E} &= \sqrt{\frac{L\gamma}{\alpha}} E, \quad \bar{F} = \sqrt{\frac{\alpha^3}{T^2 L\gamma}} F, \\ \bar{t} &= \frac{\alpha}{t_R} t, \quad \bar{\tau} = \sqrt{\frac{2\alpha}{L|\beta_2|}} \tau \end{aligned} \quad (19)$$

it can be made dimensionless:

$$\frac{\partial E}{\partial t} = F + \left[-1 - i\Delta - i\eta \frac{\partial^2}{\partial\tau^2} + i|E|^2\right] E \quad (20)$$

which is the same as equation 2 except with the added coefficient $\eta = \pm 1$ that takes the sign of β_2 . It denotes normal (+1) or anomalous (-1) dispersion. In normal dispersion, the refractive index decreases as the wavelength increases, while in anomalous dispersion, the refractive index increases as the wavelength increases. This means that in normal dispersion, shorter wavelengths (like blue light) are refracted more than longer wavelengths (like red light), while in anomalous dispersion, longer wavelengths can be refracted more than shorter ones. Also the bars have been dropped to make the equation simpler. This is the purely temporal dimensionless Lugiato-Lefever equation that we are interested in solving numerically.

2 Numerically Solving the LLE

2.1 The Split-Step Fourier Method

To solve the LLE, we use a split step method that involves treating the linear and nonlinear parts of the LLE separately. It works by evolving the nonlinear part of the solution by a step in the time domain then evolving the dispersion term by a step in the spectral domain. Although this can be done for other versions of the LLE, consider the dimensionless temporal LLE (equation 20) written as such:

$$\frac{\partial E(t, \tau)}{\partial t} = [-1 - i\Delta + i|E|^2] E - i\eta \frac{\partial^2 E}{\partial \tau^2} + F$$

where we have rearranged some of the terms to have the nonlinear part in the first bracket and the linear part is the second (dispersion) term. We assume the effects of the first term and second term happen individually. Rewriting more compactly, we have

$$\frac{\partial E}{\partial t} = (\hat{L} + \hat{N})E + F \quad (21)$$

where

$$\begin{aligned} \hat{N} &= [-1 - i\Delta + i|E|^2] \\ \hat{L} &= -i\eta \frac{\partial^2}{\partial \tau^2} \end{aligned} \quad (22)$$

The split-step Fourier method (SSFM) uses the fact that the linear and nonlinear parts of the PDE each have their own analytical solution even though equation 21 does not. This is why each part is treated separately. Starting with the linear part, we use the fact that the Fourier transform of the dispersion term does not contain a second derivative. Instead, it contains the difference between the envelope's frequency ω and the central frequency of the spectrum ω_0 .

$$\mathcal{F}\hat{L} = i\eta(\omega - \omega_0)^2 \quad (23)$$

This means that the linear step $\hat{L}E$ is best solved in the spectral domain. Since there is an analytical solution for this part, we can take a small step through t by multiplying by said solution in the spectral domain: $e^{i\eta(\omega - \omega_0)^2(dt)}$, where dt is the step size just like using first order Euler's method. The same can be done with $\hat{N}$ except there is no need to switch to the spectral domain. We multiply by the solution which is in the form: $e^{(\hat{N}+F)dt}$ [13]. Putting it all together, the SSFM algorithm used here to solve the LLE can be described by the following steps:

1. Take the Fourier transform of the solution to switch E into the spectral domain.

2. Evolve the linear part in the spectral domain by a step by multiplying by $e^{i\eta(\omega-\omega_0)^2 dt}$.
3. Inverse Fourier transform to go back into the time domain.
4. Multiply by $e^{(\hat{N}+F)dt}$ to evolve the nonlinear part.

In general, we have:

$$E_{m+1}(t + dt, \tau) = e^{(\hat{N}+F)dt} \left[\mathcal{F}^{-1} \{ e^{i\eta(\omega-\omega_0)^2 dt} \mathcal{F}(E_m(t, \tau)) \} \right] \quad (24)$$

where the subscript m denotes the current iteration in the algorithm and $m + 1$ is the next. This is an effective method to simulate nonlinear optics and an example is shown in figure 2 where we have solved equation 10 to simulate a pulse propagating through a nonlinear optical fiber. Furthermore, we can use higher order methods like

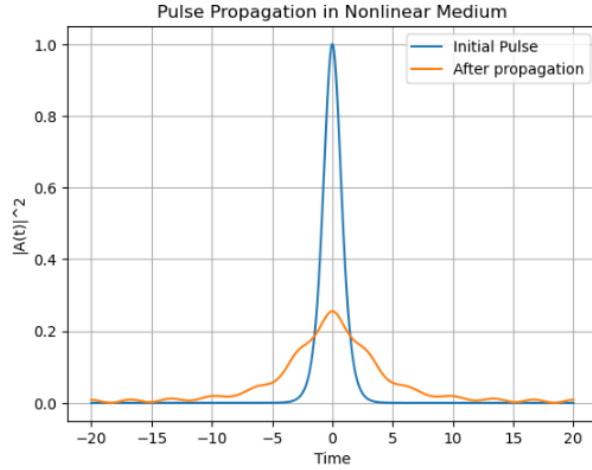


Figure 2: Plot of an initial hyperbolic secant pulse propagating through a nonlinear medium. It was generated by solving equation 10 using the split step Fourier method described in section 2 with $\beta_2 = -1$ and $\gamma = 1$.

the 2nd order Runge-Kutta when evolving the nonlinear part in step 4 for improved efficiency.

2.2 Second Order Runge-Kutta

The second order Runge-Kutta (RK) method is a numerical technique used to solve ordinary differential equations of the form:

$$\frac{dy}{dt} = f(t, y), \quad y(t_0) = y_0 \quad (25)$$

Given a step size h , the method advances from y_n to y_{n+1} using the following formulas:

$$y_{n+1} = y_n + k_2 \quad (26)$$

where

$$k_1 = hf(x_n, y_n) \quad (27)$$

$$k_2 = hf(x_n + \frac{1}{2}h, y_n + \frac{1}{2}h)$$

This method uses the slope at the midpoint of the interval (estimated using k_1) to improve the accuracy compared to the Euler method [14]. Relating back to the LLE, suppose we have computed the linear part within a step $E_L = \mathcal{F}^{-1}\{e^{i\eta(\omega-\omega_0)^2 dt}\mathcal{F}(E(t, \tau))\}$. Then to implement the 2^{nd} order RK to the nonlinear part, we have

$$E(t + dt, \tau) = E_L + k_2 \quad (28)$$

with

$$k_2 = dt \left[\hat{N} \left(E_L + \frac{k_1}{2} \right) + F \right] \quad (29)$$

$$k_1 = dt \left[\hat{N} E_L + F \right]$$

3 Regimes of Cavity Solitons in the LLE

When solving the LLE, we are interested in generated patterns known as solitons. The LLE admits localized solutions known as dissipative Kerr solitons (DKS). Generating these solitons requires balancing nonlinearity and dispersion, while also maintaining a balance between gain from the driving field and cavity loss. Having established the theoretical foundation of the LLE and the numerical methods used to solve it, we now explore some patterns these solitons can form with a Python implementation of the SSFM. All Python code and gif animations of the results can be accessed at this GitHub repository.

A typical soliton solution of the LLE takes the form of a *sech*² pulse as seen in figure 3. In experiments, a laser tuning technique is used where the CW pump laser is swept from below the cavity's resonance frequency (blue-detuned) to above its resonance frequency (red-detuned). This is known as forward tuning and often leads to multiple soliton generation in the red-detuned range [15], [10], [16]. Also worth noting is the fact that every instance of the SSFM being used in this section uses an initial perturbation in $E(t, \tau)$ in the form of a gaussian pulse. This acts as an initial guess in our iterative algorithm.

3.1 The Temporal Cavity Soliton

When the soliton is stable throughout slow time just like the system simulated in figure 3, it is known as a temporal cavity soliton (TCS). It propagates without being affected by loss and it self reinforces which is why the intensity profile and spectrum do not change much throughout slow time.

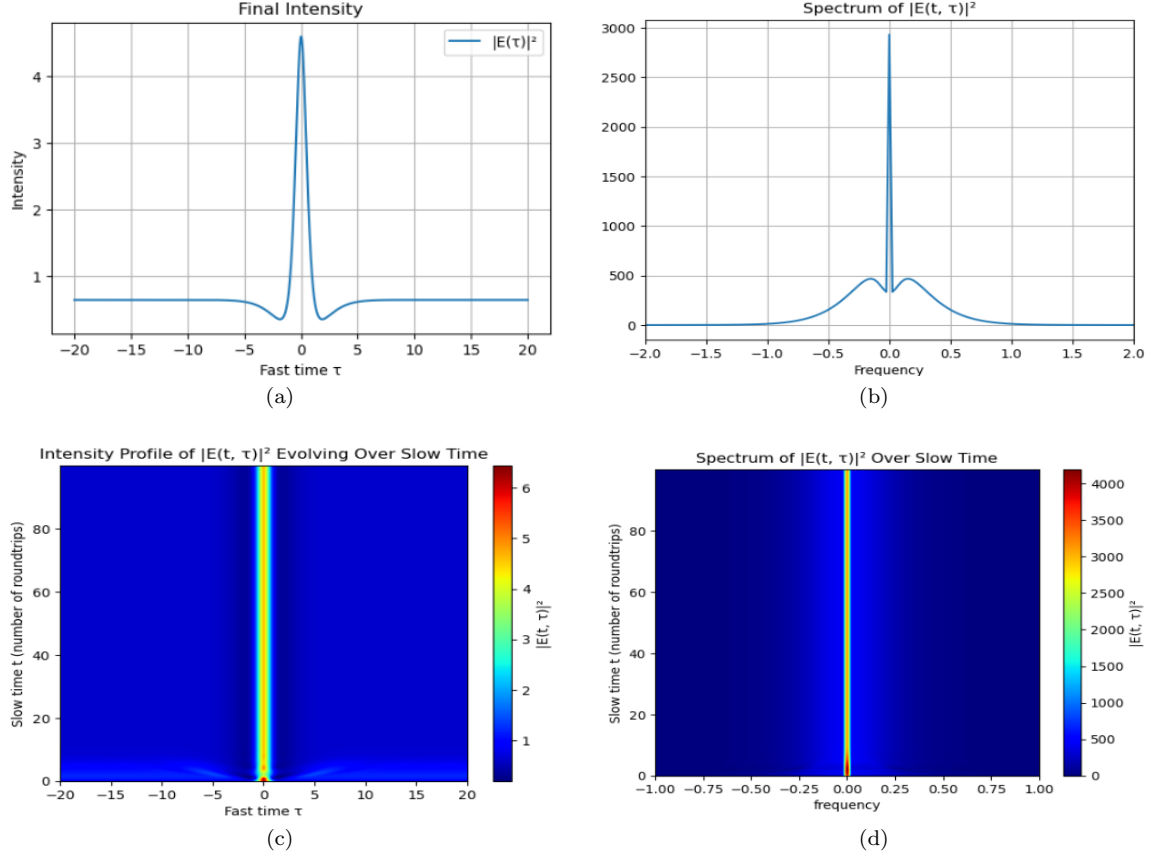


Figure 3: Stable temporal cavity soliton in the anomalous dispersion regime with parameters: $\eta = -1$, $\Delta = 2.1$, $F = \sqrt{2}$. (a) The fast-time intensity profile $|E(\tau)|^2$ after 100 roundtrips matches the expected sech^2 shape of a dissipative Kerr soliton. (b) The corresponding spectrum exhibits a sharp comb line. (c) Evolution of $|E(t, \tau)|^2$ over slow time shows the soliton maintaining its shape and amplitude with minimal drift, indicating long-term stability. (d) The spectrum remains unchanged throughout, consistent with a stationary soliton producing a stable frequency comb.

3.2 Multiple Solitons

In the multiple soliton regime, more than one temporal cavity soliton circulates in the resonator simultaneously. The separation between solitons in fast time can be irregular and their relative phases may drift slowly over slow time. Interactions through the CW input in the background can cause solitons to move relative to one another even if they are far apart in the fast time domain. Over many roundtrips, they may collide, merge, or disappear [17], [18], [19]. Despite not always being periodic, the comb lines remain largely coherent which makes this regime relevant for applications that do not require uniform spacing but benefit from multiple pulses circulating in the cavity. We can simulate how these solitons form by simulating the optical field's intensity as it evolves through slow time t . In figure 4b multiple solitons are depicted that do not have equal spacing. In (d), we can see a multiple soliton regime for most of the detuning scan where the solitons drift apart as detuning rises. Such states often form when the system enters a detuning range that supports soliton formation from a turbulent or instable regime [20].

3.3 Soliton Crystal

Soliton crystals are essentially regularly spaced arrays of TCSs inside a Kerr resonator. They can be thought of as a multiple soliton regime where the solitons are arranged in a perfectly periodic pattern in the fast time domain. Additionally, these solitons are phase-locked. That is, the carrier waves within these solitons maintain a constant relative phase with each other and the pump field. This ensures a stable array of pulses in time and an evenly spaced comb spectrum in frequency. In a soliton crystal, the solitons settle into fixed positions separated by an integer multiple of the roundtrip time. Soliton crystals are also characterized by sidebands in the spectrum around the sharp comb as seen in the spectrum of 5a. The sidebands appear because the solitons are not only identical, but also arranged in an evenly spaced lattice in time. Typically, to generate a soliton crystal we have to reduce the input field compared to when we want to generate multiple solitons [21]. We can also infer from figure 4d which ranges of detuning seem to generate evenly spaced solitons. For $F = \sqrt{3}$, $\Delta \approx 1$ looks promising. Thus, reducing the input field intensity and setting the detuning to $\Delta = 1.2$ generates the state seen in figure 5.

Figures 4b and 5b both depict regimes in which multiple temporal cavity solitons circulate in the resonator. However, their spatial organization and spectral signatures differ. The solitons in figure 4b may be stable even after 80 roundtrips but they do not generate the frequency profile with sharp sidebands like those in 5a and 5b. Since the solitons are not all equally spaced, we see a comb spectrum with a relatively smooth envelope (4a). Meanwhile, the soliton crystal state (figure 5)

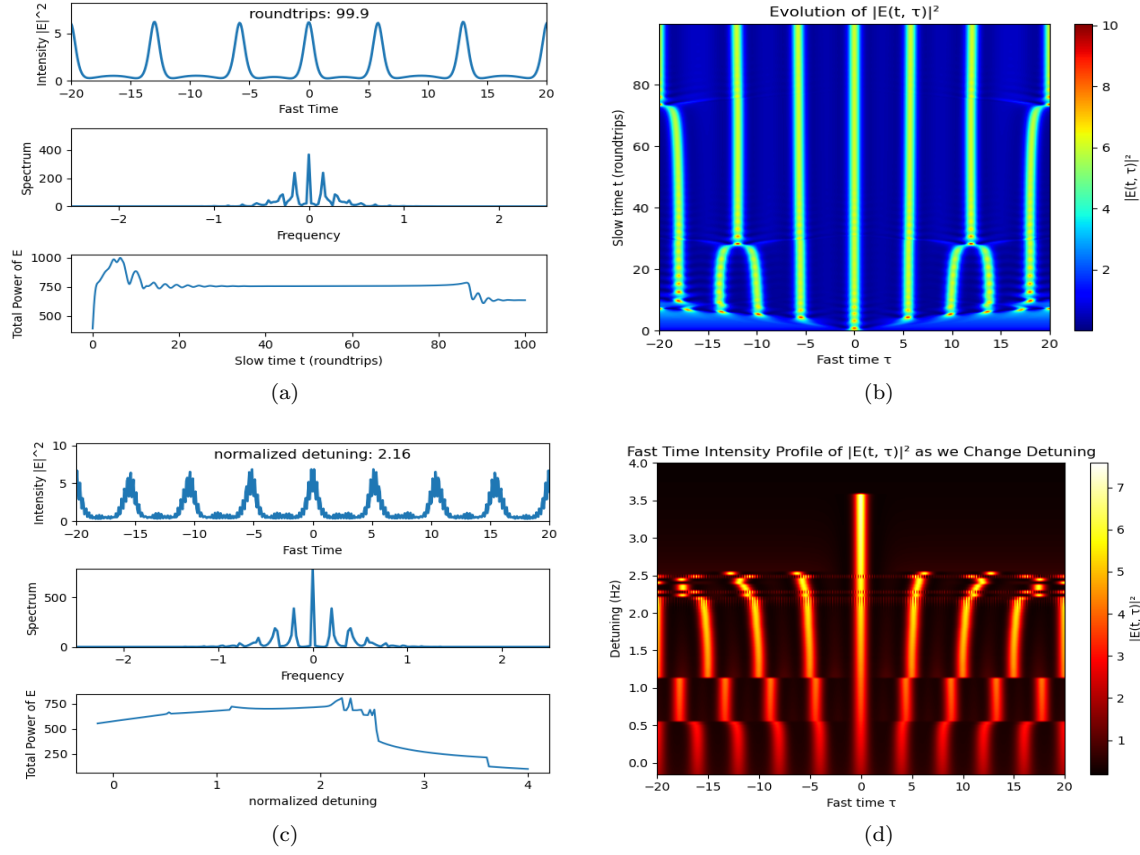


Figure 4: The figure shows the multiple soliton regime with $\eta = -1$, $F = \sqrt{3}$ and $\Delta = 2.45$. (a) shows the evolution of $|E(t, \tau)|^2$, its spectrum over 100 trips around the ring and the total intracavity power. (b) Illustrates uneven soliton spacing in fast time, a signature of weak long-range interactions through the continuous-wave background. (c) shows a detuning scan where (Δ ranges from -0.15 to 4) and changes after 5 roundtrips. In the detuning scan of (d), several multi-soliton regimes are produced where the solitons drift until falling into a single soliton regime.

is characterized by perfectly periodic pulse spacing, forming a stable time domain lattice. This periodicity produces several sharp combs in the spectral domain.

3.4 Chaotic Regime

There even exists a chaotic regime where the intracavity field fails to settle into a stationary or periodic pattern, instead exhibiting irregular evolution in both fast and slow time. The system still exhibit Kerr nonlinearity and dispersion but typically have input field and detuning values that do not form solitons. Instead, the localized

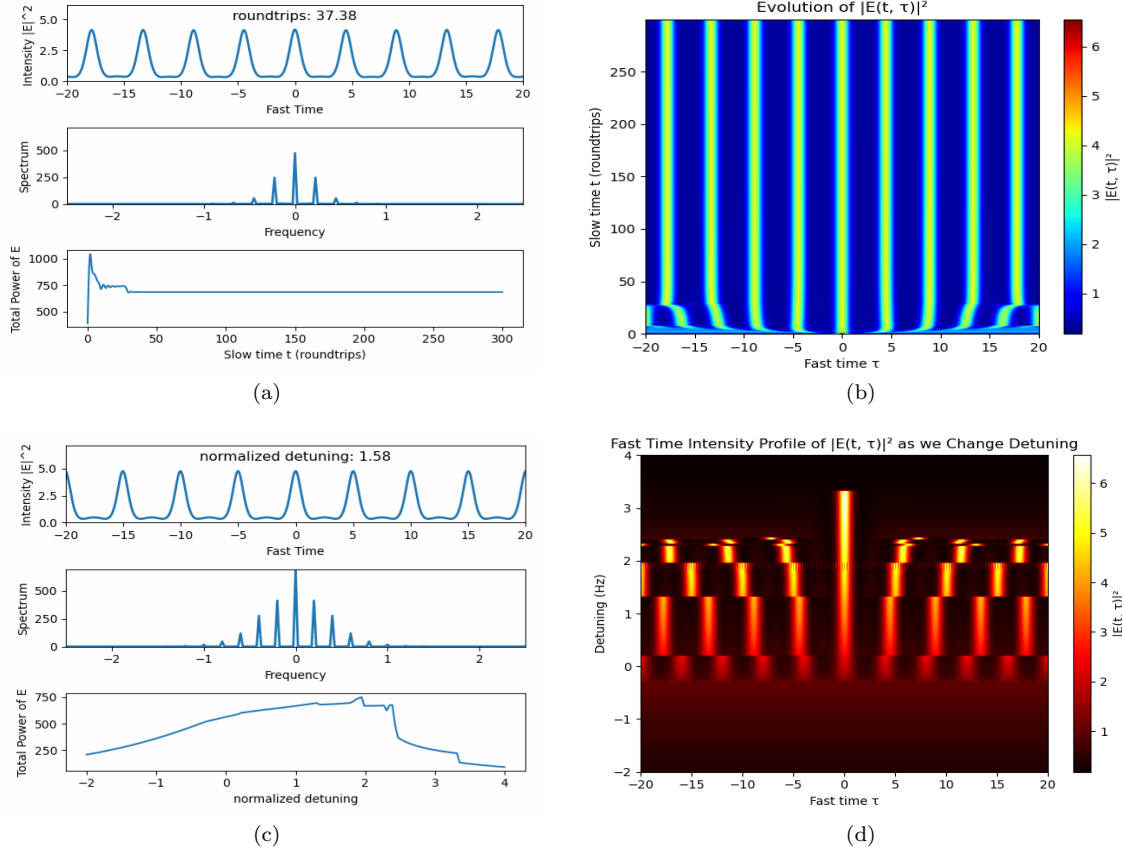


Figure 5: Soliton crystal formation with $\eta = -1$, $F = 1.4$, detuning swept from $\Delta = -1$ to 2.5 where the detuning changes after every 10 roundtrips. (a) Crystal soliton regime with $\eta = -1$, $F = \sqrt{2.75}$ and $\Delta = 1.2$ over 300 roundtrips of the ring. The evenly spaced pulses in $|E(t, \tau)|^2$ reveal phase-locked solitons forming a time domain lattice. There are also sharp sidebands next to the frequency comb. (b) The fast-time profile confirms periodicity, in contrast to the unequal spacing in the multiple soliton regime. (c) The spectrum shows sharp comb-like lines with sidebands inside the soliton crystal regimes. (d) Detuning scan with $\Delta = -2$ to $\Delta = 4$ where the detuning changes every 5 roundtrips. The crystal state exists from $\Delta \approx 0.15$ to 1.4 and from $\Delta \approx 1.4$ to 2.

structures interact in a turbulent manner.

In fast time, the pulse profiles fluctuate in both amplitude and position from roundtrip to roundtrip, producing no regular spacing or consistent envelope shape. In the frequency domain, the spectrum is typically broad but lacks the well-defined modulation sidebands characteristic of a soliton crystal or the smooth envelope of a single-soliton state. Although comb lines may still be discernible, their amplitudes and spacings vary significantly over time as seen in 6. The system in 6a and 6b

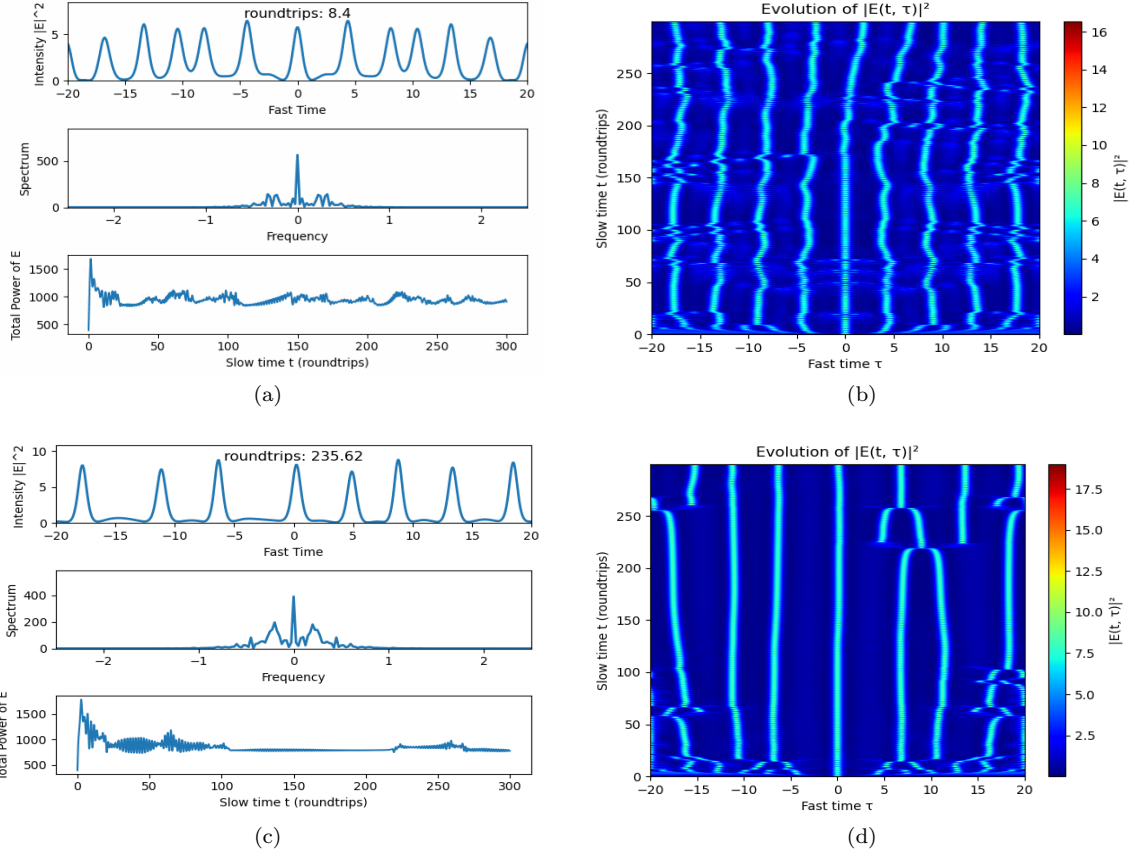


Figure 6: (a) System with $\eta = -1$, $\Delta = 2$ and $F = \sqrt{5}$ over 300 roundtrips. (b) Slow time evolution of the system in (a). (c) System with $\eta = -1$, $\Delta = 3$ and $F = \sqrt{5}$ over 300 roundtrips. (d) Slow time evolution of the system in (c).

demonstrates the varying amplitudes mentioned previously. Meanwhile the system in 6c and 6d shows uneven spacing even with new structures appearing and disappearing throughout slow time as well as the variation in amplitude. This system experiences chaotic sections in slow time before briefly settling into a multiple soliton state at around 110 roundtrips (6d).

4 Conclusion

This work examined the temporal Lugiato–Lefever Equation as a model for intracavity field dynamics in Kerr optical resonators. Starting from the nonlinear Schrödinger equation, the LLE was derived by incorporating Kerr nonlinearity, group-velocity dispersion, cavity driving and loss. Then it was transformed to be dimensionless. Numerical simulations using a Split-Step Fourier Method with a second-order Runge–Kutta nonlinear step reproduced different soliton regimes, including single temporal cavity solitons, multiple soliton states, and soliton crystals. The results highlight how variations in detuning and pump strength govern the transition between these states. In particular, the soliton crystal regime produced highly regular, phase locked frequency combs with strong sideband modulation, while the multiple-soliton regime exhibited irregular spacing and less pronounced modulation. These findings demonstrate the usefulness of the LLE as a tool for understanding Kerr frequency comb generation. Future research could include modifying the simulation to solve the dimensional temporal LLE (equation 18) or the dimensional spatial LLE. This would be useful for simulating physical microresonators.

5 Appendices

This section contains all the Python scripts used to simulate nonlinear fiber optics and the evolution of the intracavity field in a Kerr medium resonator. All the code and animations can be found in this GitHub repository.

A Python implementation of the SSFM to solve the NLSE

```
import numpy as np
import matplotlib.pyplot as plt
nt = 1024          # Number of time points
tmax = 20          # Max time window
dt = 2 * tmax / nt # Time step
t = np.linspace(-tmax, tmax, nt)

dz = 0.01          # Step size in z
nz = 1000          # Number of z steps

# Frequency domain
w = 2 * np.pi * np.fft.fftfreq(nt, d=dt)

# Physical parameters
# Dispersion coefficient (beta_2 < 0 for anomalous dispersion)
beta2 = -1
beta3 = 0          # Distorts pulse
gamma = 1          # Nonlinearity coefficient

# Initial condition: Sech pulse
A0 = 1/np.cosh(t)

# Allocate array
A = A0.copy()

# Precompute some terms
dispersion = np.exp(0.5j * beta2 * w**2 * dz
+ (1j/6) * beta3 * w**3 * dz )
#for dispersion (taylor series...)
#SSFM
for _ in range(nz):
```

```

    # Linear step (half-step)
    A = np.fft.ifft(np.fft.fft(A) * dispersion)
    # Nonlinear step
    A = np.exp(1j * gamma * np.abs(A)**2 * dz) * A
    # Linear step (half-step)
    A = np.fft.ifft(np.fft.fft(A) * dispersion)
# Plots
plt.figure()
plt.plot(t, np.abs(A0)**2, label='Initial Pulse')
plt.plot(t, np.abs(A)**2, label='After propagation')
plt.xlabel('Time')
plt.ylabel('|A(t)|^2')
plt.legend()
plt.title('Pulse Propagation in Nonlinear Medium')
plt.grid()
plt.show()

```

B Python implementation of the SSFM to solve the dimensionless LLE

```

import numpy as np
import matplotlib.pyplot as plt
from numpy.fft import fft, ifft, fftfreq, fftshift
import matplotlib.animation as animation

class LLE_sim:
    def __init__(self, eta, detuning, F, T, roundtrips, N):
        self.eta = eta
        self.detuning = detuning
        self.F = F #driving
        self.T = T #fast time window
        self.roundtrips = roundtrips

        # Fast time grid
        self.N = N #fast time grid points
        self.tau = np.linspace(-self.T/2, self.T/2, self.N)
        self.dtau = self.tau[1] - self.tau[0]

        # Frequency grid

```

```

self.omega = 2 * np.pi * fftfreq(self.N, d=self.dtau)

# Save evolution
self.dt = 0.01 #step size
self.n = int(self.roundtrips/self.dt)
self.E_evolution =
np.zeros((self.n, self.N), dtype=complex)
self.t_vals = np.arange(self.n)*self.dt
def nonlinear_step(self, A):
    return (- 1 - 1j * self.detuning + 1j * np.abs(A)**2) * A

def ssfm(self): #solve LLE
    dt = self.dt #slow time step
    E = 0.75 + 1.8 * np.exp(-self.tau**2 / (0.75)) #initial pulse
    D = np.exp(1j * self.eta * self.omega**2 * dt)
    for step in range(self.n):
        #linear step
        E = fft(E)
        E = ifft(E * D)

        #nl step with detuning, loss, driving (using rk2)
        k1 = dt * (self.nonlinear_step(E) + self.F)
        k2 = dt * (self.nonlinear_step(E + k1/2) +self.F)
        E = (E + k2)
        #save intensity profile for slow time plots
        self.E_evolution[step] = E
    return E

def plot_peak_power(self):

    peak_power = np.max(np.abs(self.E_evolution)**2, axis=1)
    plt.figure()
    plt.title('Peak Power of E')
    plt.plot(self.t_vals, peak_power)
    plt.xlabel('Slow time t (s)')
    plt.ylabel('W')
    plt.grid()

def get_spectrum(self, sol):

```

```

    spectrum = np.fft.fft(sol)
    frequencies = np.fft.fftfreq(len(sol), self.dtau)
    spectrum_magnitude = np.abs(np.fft.fftshift(spectrum))
    frequencies_shifted = np.fft.fftshift(frequencies)
    return frequencies_shifted, spectrum_magnitude

def plot_slow_time_evolution(self):
    self.ssfm()
    if self.eta == -1:
        print("=", self.eta, " (Anomalous dispersion)")
    elif self.eta == 1:
        print("=", self.eta, " (Normal dispersion)")

    print("detuning =", self.detuning)
    print("F =", self.F)

    # Plot space-time diagram of intensity
    plt.figure()

    plt.imshow(np.abs(self.E_evolution)**2, aspect='auto',
        extent=[self.tau[0], self.tau[-1],
        , self.t_vals[0], self.t_vals[-1]], cmap='jet' , origin='lower')

    plt.xlabel('Fast time ')
    plt.ylabel('Slow time t (roundtrips)')
    plt.title('Evolution of  $|E(t, )|^2$ ')
    plt.colorbar(label='|E(t, )|^2')
    plt.show()
def animate_slowtime(self):
    # animate the above plot
    fig, axs = plt.subplots(3, 1)
    fast_time = self.tau
    line, = axs[0].plot([], [], lw=2)
    line1, = axs[1].plot([], [], lw=2)

    axs[0].set_xlim(fast_time[0], fast_time[-1])
    axs[0].set_ylim(0, np.max(np.abs(self.E_evolution)**2) * 1.5)
    axs[0].set_xlabel('Fast Time')
    axs[0].set_ylabel('Intensity  $|E|^2$  ')

```

```

    axs[1].set_xlim(-2.5,2.5)
    axs[1].set_xlabel('Frequency')
    axs[1].set_ylabel('Spectrum')

    def init():
        line.set_data([], [])
        line1.set_data([], [])
        return line, line1

    def update(frame):
        axs[0].set_ylim(0, np.max(np.abs(self.E_evolution[frame])**2) + 2)
        line.set_data(fast_time, np.abs(self.E_evolution[frame])**2)
        axs[0].set_title(f'roundtrips: {frame/100}', y = 1.0, pad=-14)
        line1.set_data(self.get_spectrum(self.E_evolution[frame])[0]
            ,self.get_spectrum(self.E_evolution[frame])[1] )
        axs[1].set_xlim(-2.5,2.5)
        axs[1].set_ylim(0, np.max(self.get_spectrum
            (self.E_evolution[frame])[1]) * 1.5)
        return line,line1

    frames = range(0,self.n, 21)
    self.ani = animation.FuncAnimation(fig, update, frames=frames
    , init_func=init, blit=False, interval=0.001)

    total_power = []

    for i in range(len(self.t_vals)):
        total_power.append(sum(np.abs(self.E_evolution[i])**2))

    axs[2].plot(self.t_vals, total_power)
    axs[2].set(xlabel =
    'Slow time t (roundtrips)', ylabel = 'Total Power of E' )
    plt.tight_layout()

    self.ani.save('slow_time_evolution.gif', writer = 'pillow')

def scan_detuning(self, delta_start, delta_end, delta_steps, x):
    self.roundtrips = x # change the detuning after x roundtrips

```

```

delta_values = np.linspace(delta_start,delta_end, delta_steps)
E_profiles = []
for delta in delta_values:
    self.detuning = delta
    E_profiles.append(np.abs(self.ssfm())**2)

#create plot of intracavity power and intensity profile as we scan detuning
fig, axs = plt.subplots(3, 1)
fast_time = self.tau
line, = axs[0].plot([], [], lw=2)
line1, = axs[1].plot([], [], lw=2)

axs[0].set_xlim(fast_time[0], fast_time[-1])
axs[0].set_ylim(0, np.max(E_profiles) * 1.5)
axs[0].set_xlabel('Fast Time')
axs[0].set_ylabel('Intensity  $|E|^2$ ')
axs[1].set_xlim(-2.5, 2.5)
axs[1].set_xlabel('Frequency')
axs[1].set_ylabel('Spectrum')

def init():
    line.set_data([], [])
    line1.set_data([], [])
    return line,line1

def update(frame):
    line.set_data(fast_time, (E_profiles[frame]))
    line1.set_data(self.get_spectrum(E_profiles[frame])[0]
    ,self.get_spectrum(E_profiles[frame])[1] )
    axs[1].set_xlim(-2.5, 2.5)
    axs[1].set_ylim(0,
    np.max(self.get_spectrum(E_profiles[frame])[1]) +2)
    axs[0].set_ylim(0, np.max(E_profiles[frame]) * 1.5)
    axs[0].set_title(f'normalized detuning:
    {round(delta_values[frame],2)}', y = 1.0, pad=-14)
    return line,line1

self.ani = animation.FuncAnimation(fig, update
, frames=len(E_profiles), init_func=init, blit=False, interval=100)
total_power = []

```

```

for i in range(len(delta_values)):
    total_power.append(sum((E_profiles[i])))

axs[2].plot(delta_values, total_power)
axs[2].set(xlabel = 'normalized detuning', ylabel = 'Total Power of E' )
plt.tight_layout()
self.ani.save('Multiple-soliton-detuning-scan.gif', writer = 'pillow')

# Fast time intensity profile
plt.figure()
plt.title('Fast Time Intensity Profile of  $|E(t, )|^2$  as we Change Detuning')
plt.imshow(E_profiles, extent=[self.tau[0], self.tau[-1],
delta_values[0], delta_values[-1]],
aspect='auto', cmap='hot', origin='lower')
plt.xlabel('Fast time ')
plt.ylabel('Detuning (Hz)')
plt.colorbar(label = ' $|E(t, )|^2$ ')
plt.tight_layout()
plt.show()
sol = LLE_sim(
    eta = -1,
    detuning = 1.2,
    F = np.sqrt(2.75),
    T = 40,
    roundtrips = 300,
    N = 2**9
)

###
sol.plot_slow_time_evolution()
###
sol.animate_slowtime()
###
sol.scan_detuning(-2,4,150,5)

```


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